

Mason Bane

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Summary: Computer Engineering student with hands-on hardware experience in circuit design, PCB prototyping, and microcontroller systems. Proficient in hardware analysis tools and low-level programming for complete hardware/software integration. Brings research background and analytical skills to hardware design challenges.

EDUCATION

Tarleton State University

Stephenville, TX

Bachelor of Science in Computer Science, Concentration in Computer Engineering Aug. 2022 – May 2026 (Expected)

Minor in Mathematics

GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Hill College

Hillsboro, TX

Associate of Arts in Liberal Arts

Aug. 2019 – Sept. 2023

EXPERIENCE

Undergraduate Research Assistant - Systems & Performance

May 2024 – Present

Tarleton State University

Stephenville, TX

- Developed high-performance simulations in C/C++, applying parallel algorithms on Linux clusters to accelerate computation.
- Worked closely with hardware teams to profile performance and align software with system constraints for reliable runtimes.
- Documented deployment and debugging procedures to support cross-functional teams.

Undergraduate Technology Specialist - HPC Lab Manager

Aug. 2024 – Present

Tarleton State University

Stephenville, TX

- Managed 15+ Linux systems and associated hardware, performing diagnostics, repairs, and component replacements to maintain uptime.
- Monitored thermal, memory, and storage health to prevent failures and extend equipment life.
- Maintained organized documentation of hardware configurations and maintenance procedures for repeatability.

PROJECTS

Secure IoT Sensor Node with Hardware TrustZone | C, STM32CubeIDE/HAL, I2C/SPI

Dec. 2025 – Present

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

Analog Pink Noise Generator Circuit | LTSpice, Python, MATLAB, Breadboarding

Aug. 2025 – Dec. 2025

- Designed and prototyped a BJT-based pink noise generator with active filter stage using LTSpice for simulation and breadboarding for physical implementation.
- Developed Python scripts to quantitatively compare simulation results with an ideal pink noise spectrum, guiding component selection and filter tuning for improved accuracy.
- Built a physical demonstration circuit with audio output, successfully verifying simulation models against real-world performance.
- Collaborated in a two-person team to present comprehensive design methodology, results, and data-driven decision process to peers and professors.

TECHNICAL SKILLS

Hardware Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi, ARM Cortex-M

Firmware Development: C, C++, ARM Assembly, I²C, SPI, UART, STM32CubeIDE/HAL, RTOS

Circuit Design & Tools: LTSpice, Analog/Digital Circuit Design, PCB Prototyping, Oscilloscope, Multimeter

Development Tools: Git, Linux, CMake/Make, GDB, Python, Bash, VS Code

Simulation & Research: MATLAB, CUDA, OpenGL, Blender